Diagonal Redundancy, Not Sparsity, in the Regression Block: A Preregistered Test and the Limits of Pair Subsampling

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Abstract

This paper's preregistered test evaluated diagonal redundancy, not sparsity. The Regression Block of Goulet Coulombe (2026) replaces a transformer's attention sublayer with a regression on degree-2 polynomial cross-features, matching the FT-Transformer on eight tabular benchmarks; its stated binding constraint is the quadratic construction of that cross-feature map. We preregistered a strict drop-in that keeps the architecture, PCA (principal component analysis) compression, ridge warm start (warm-starting the ridge regression head), and training fixed but replaces the full off-diagonal pair set with a fixed random subset of q pairs: $q = 4P$ (primary) or $q = 2P$ (exploratory), where P is the number of input features.

Up front: at every benchmark dimension ($P \leq 9$), $q = 4P$ contains the full off-diagonal pair set, so the primary arm keeps every off-diagonal pair and drops only the P redundant diagonal self-products — nothing random happens in this arm. The preregistered test therefore confirms the diagonal ablation is free on 7 of 8 datasets — a confirmation, not a severe test, since the deterministic arm (nothing random happens) can only corroborate the a priori argument below: the paired mean difference in out-of-sample R^2 (held-out variance explained; ΔR^2, averaged over the 40 dataset×split cells) is -0.0003 (95% confidence interval, CI, $[-0.0049, +0.0043]$), inside the -0.01 non-inferiority margin (i.e., not worse than the full map by more than the pre-set tolerance) under the pooled test (the primary decision rule; several per-dataset CIs cross the margin); the dataset-level check (averaging splits within each of the 8 datasets, $n = 8$) agrees. The one exception is Kin8nm, whose per-dataset CI sits entirely below zero (-0.0035) — plausibly noise under multiple comparisons, but enough to keep "universally redundant" off the table. The margin is wider than the source paper's own published $+0.007$ advantage over the FT-Transformer, so a non-inferior verdict can surrender about 1.4× that edge; the pooled CI also clears a stricter -0.005 margin.

The $q = 2P$ arm — exploratory, the only genuine sparsity test at these dimensions — loses -0.0043 pooled (CI $[-0.0095, +0.0008]$), with systematically negative losses concentrated on Kin8nm and Protein. The dataset-level check (2/8 wins, $p = 0.29$; CI $[-0.0108, +0.0022]$) marginally crosses the -0.01 margin, so the genuine-sparsity verdict is descriptive only; the pooled sign-test ($p = 0.002$ across correlated splits) is reported for completeness — a sign test counts how many cells favor SparsePair-2P versus the baseline. The source's own PCA→50 trimming (retaining 50 principal components) ties DiagonalFree-4P on accuracy ($+0.0002$, CI $[-0.0031, +0.0036]$) while being cheaper in parameters and wall time: at $P \leq 9$ the source's trimming dominates on every measured cost.

In a post-hoc, underpowered follow-up on the CPU_act and Bank32nh tabular benchmarks ($P = 21$ and $P = 32$) — the source's Appendix F regime, where $q = 4P$ keeps only 40% and 26% of the off-diagonal pairs — DiagonalFree-4P is descriptively close ($+0.0006$ and -0.0039 mean ΔR^2; Bank32nh CI $[-0.0160, +0.0082]$) — "descriptively close" meaning the point estimates are near the baseline with no statistical match claimed — and the constructed feature width falls 63% and 75% (14,912→5,504; 33,920→8,320). We measured no wall-clock or memory benefit at any P: the result is width reduction at $P \geq 21$ plus analytic $O(P^2) \to O(P)$ scaling, not a faster model. An independent NumPy reimplementation —

written from scratch, following the published pipeline of the authors' MIT-licensed reference implementation, adapted with attribution — reproduces the paper's Regression Block column to 0.001 on the mean.

1 Introduction

Goulet Coulombe (2026) shows that ordinary least squares predictions can be rewritten as the output of a restricted attention module, and builds on the equivalence in both directions. In the reverse direction, the paper introduces the *Regression Block*: a transformer block in which the softmax attention sublayer is replaced with explicit regression on degree-2 polynomial cross-features. On eight UCI/OpenML tabular regression benchmarks, the single-block Regression Block roughly matches the three-block FT-Transformer (Gorishniy et al., 2021) (mean R^2 0.786 vs. 0.779 in the source's Table 1) at roughly 30% of the parameters (30K vs. 100K) and a third of the depth. The ablations identify the polynomial expansion as the most important ingredient by far.

The paper is candid about the cost. The Limitations section states that the cross-features "grow quadratically with the number of inputs, which becomes the binding constraint" beyond the benchmark dimensions: at $P = 81$ inputs, the author's runs required an A100 GPU with small batches, and one run timed out. The paper does report a cheaper variant, the trimmed configuration of Appendix E (two mixing sublayers, no feed-forward sublayer, 50 principal components). But that variant cuts the PCA basis width and drops the feed-forward sublayer rather than the $O(P^2)$ construction, is presented as an ablation rather than a proposed fix, and loses 0.010–0.036 relative to the full block at $P = 21$–48 (Appendix F). It proposes no remedy for the construction cost itself.

This paper proposes and preregisters a test of the simplest possible remedy. The quadratic map is highly redundant: the paper's own ablations show that 50 principal components retain more than 99.99% of the polynomial basis's variance (mean R^2 of 0.785 at 200 components against 0.783 at 50 in this paper's independent rerun). If the interaction content is effectively low-dimensional, a sparse random subset of the pairwise products should preserve it. We replace the full set of $P(P+1)/2$ pairs with a fixed random subset of q off-diagonal pairs, with $q = 4P$ as the preregistered primary variant and $q = 2P$ as an exploratory one. Everything else — tokenizer, PCA width, ridge warm start, optimizer, seeds, splits — is held fixed, so the comparison isolates the feature map. Feature *construction* falls from $O(P^2 d)$ to $O(Pd)$ while the trainable parameter count is unchanged.

One qualification belongs in the introduction, not the fine print. At the benchmark dimensions ($P \leq 9$), $q = 4P$ exceeds the full off-diagonal pair count on every dataset: the preregistered primary arm retains every off-diagonal pair and drops only the P diagonal self-products, which duplicate the squared terms already present in the per-token block (defined in Eq. (1) below). Read honestly, the primary test is therefore an ablation establishing that the diagonal is redundant. The $O(P)$ construction scaling is by design, genuinely exercised only at larger P (Section 5 quantifies this per dataset). The $q = 2P$ arm is the true sparsity test at these dimensions.

Our contributions: (i) a tested drop-in that subsamples the Regression Block's quadratic cross-feature map to $O(P)$ pair counts, as a strict architectural replacement that changes no parameter and no training choice; (ii) a preregistered non-inferiority test on the paper's full benchmark suite (8 datasets $\times$ 5 splits, 120 trained models), with an independent reimplementation that reproduces the paper's numbers; (iii) an honest accounting of what each arm actually tested — diagonal redundancy for the primary arm, genuine $\sim$50% sparsity for the exploratory one — and of where aggressive sparsity starts to cost accuracy.

2 Background: the Regression Block and its bottleneck

In plain terms, the Regression Block works as follows. Each input feature gets its own small vector (a token), the block forms all pairwise products between feature values, compresses this large product set with a frozen PCA, and feeds the result through a tiny neural network that predicts the target. It is a transformer with its attention sublayer swapped out for this explicit quadratic regression. We summarize

the construction of Goulet Coulombe (2026) as implemented in the paper's released code (MIT-licensed reference implementation, reimplemented here independently in NumPy; details in Appendix B). Each of the P input features becomes a token via a shared linear map $\mathbb{R} \to \mathbb{R}^d$ ($d = 64$) plus a learned per-feature positional embedding, giving $e \in \mathbb{R}^{P \times d}$. The mixing sublayer maps each token to degree-2 cross-features

$$\phi(e) = \big[\, e_p,\ e_p^2 - 1,\ \{\, e_i \odot e_j : 1 \le i \le j \le P \,\} \text{ for each token } p \big], \tag{1}$$

of width $2d + \frac{P(P+1)}{2} d$: linear and squared terms per token plus all pairwise elementwise products across feature positions (the pair vector is identical across tokens). These are compressed by a PCA to 200 components — fit once on training tokens from a randomly initialized tokenizer, then frozen — passed through a linear map $W : \mathbb{R}^{200} \to \mathbb{R}^{64}$, and added to the residual stream with post-norm LayerNorm. A position-wise FFN ($64 \to 128 \to 64$, ReLU, dropout 0.1), a second residual + LayerNorm, mean-pooling over tokens, and a linear head complete the single block. Before training, the first output channel of W and the head are warm-started from a ridge regression ($\lambda = 1$) of the pooled PCA features on the training target. Training uses Adam (Kingma & Ba, 2015) (10^{-3}, weight decay 10^{-4}), batch 256, up to 200 epochs with early stopping (patience 10 on a 15% validation split) and gradient clipping at norm 1. The paragraph above fixes the exact recipe for reproducibility; for the rest of the paper only the plain-language picture matters: tokens, pairwise products, PCA compression, prediction.

The bottleneck is Eq. (1): constructing the pair products costs $O(P^2 d)$ per batch, and the width itself is $O(P^2 d)$ before PCA — the binding constraint the paper reports at higher input dimension.

3 Sparse random-pair cross-features

The *SparsePair* mixer keeps every element of Section 2 fixed and replaces the pair set $\{i \le j\}$ in Eq. (1) with a fixed subset $\mathcal{S}$ of off-diagonal pairs, $|\mathcal{S}| = q$, drawn once per dataset split from a seeded RNG:

$$\phi_{\mathcal{S}}(e) = \big[\, e_p,\ e_p^2 - 1,\ \{\, e_i \odot e_j : (i,j) \in \mathcal{S} \,\} \text{ for each token } p \big]. \tag{2}$$

We test $q = 4P$ (primary, preregistered) and $q = 2P$ (exploratory), labeled DiagonalFree-4P and SparsePair-2P below — the 4P label names what the arm actually does at benchmark dimensions (drop the diagonal), not the subsampling the $q = 2P$ arm performs. Diagonal pairs are excluded because the squared terms e_p^2 already appear in the per-token block. The PCA still outputs 200 components, so the trainable parameter count is identical to the full model; only the constructed width changes ($2d + qd$ vs. $2d + \frac{P(P+1)}{2} d$). The pair subset is drawn from a dedicated RNG stream so that initialization, validation split, batch order, and dropout masks are bit-identical to the baseline on each split — the paired comparison isolates the map.

The design is deliberately minimal: it tests whether the dense pair set is needed, not whether a different nonlinearity would do better. Random subsampling of pairwise products has precedent as a feature-engineering heuristic — TabPFN, e.g., generates second-order features from up to 50 randomly selected feature pairs as an option (Hollmann et al., 2025). What has not been tested is its use as a drop-in replacement for the Regression Block's quadratic map under a preregistered protocol.

A mechanical fact that shapes the reading of every result below: the preregistration specifies $q = \min(4P, \#\text{off-diagonal pairs})$, and at every benchmark dimension $4P$ exceeds the off-diagonal count. The primary arm is therefore degenerate in one respect — on all eight datasets it retains *every* off-diagonal pair, and the "random subset" is the full set with probability one. Nothing random happens in the primary arm; the randomness bites only in the $q = 2P$ arm (and at larger P, Section 5). This does not deviate from the preregistration, which fixed the rule rather than the outcome, but it is the correct reading of what the primary test measures: an ablation of the P diagonal self-products.

A short a priori argument explains why parity was expected. The pair vector is identical across tokens (Section 2), and each dropped diagonal column $e_i \odot e_i$ lies in the column span of the retained per-token $[e_p,\ e_p^2 - 1]$ terms: per-(p,k) coefficients reconstruct the diagonal from the linear and squared terms generically. Dropping the diagonal therefore leaves the data matrix's column space exactly unchanged before the PCA (up to the mean-centering the PCA fit applies). The PCA is fit on the reduced map, so a deficit can still

arise downstream — and on Kin8nm it does (-0.0035, CI entirely below zero) — which localizes any real effect to the nonlinear downstream (PCA truncation, LayerNorm, FFN) rather than to the map itself. This constrains what a dataset-specific explanation of the Kin8nm exception would have to look like, alongside the multiple-comparisons reading offered in Section 7.

4 Preregistered protocol

We timestamped the full preregistration (hypothesis, methods, margin, and decision rule) before any experiment ran and summarized it in Appendix A; we report no deviations. Datasets: the paper's eight Table 1 benchmarks (California, Yacht, Energy, Concrete, Airfoil, Abalone, Kin8nm, Protein) from UCI/OpenML (Vanschoren et al., 2014), with larger ones capped at $N = 5000$ by subsampling with seed 42, exactly as in the paper. No dataset was excluded (all have $P \leq 9$, so the full quadratic baseline is CPU-feasible — the fair setting for the comparison). Protocol: five random 80/20 splits per dataset (seeds $42 + 1000k$), features and targets standardized on the training split (the revised paper's target-standardized convention), one fixed configuration per method, metric out-of-sample R^2. Forty dataset×split cells per method; three methods (full baseline, DiagonalFree-4P, SparsePair-2P); 120 trained models, CPU-only, seeds fixed and recorded.

Preregistered decision rule. Let $d = R^2(\text{DiagonalFree-4P}) - R^2(\text{full})$ over the 40 paired cells. Non-inferiority holds iff the lower bound of the two-sided 95% CI of the mean of d (paired t, 39 df) exceeds -0.01. The 0.01 margin is roughly the source paper's own published advantage of the Regression Block over the FT-Transformer (0.786 vs. 0.779 in the source's Table 1; pooled paired difference $+0.007$, $p = 0.009$), and the 0.01 margin is therefore *wider* than the published advantage it is anchored to: declaring non-inferiority at this margin permits the sparse map to surrender up to about 1.4× that $+0.007$ edge. For the record: the timestamped preregistration anchored the margin to the source's "within 0.005–0.01" language for its trimmed configurations; the $+0.007$ anchoring above is the manuscript's post-hoc rewording of the same margin value, disclosed here. Non-inferiority here does not preserve the Regression Block's claimed advantage over the FT-Transformer; it tolerates surrendering more than all of it. Because the five splits within a dataset share training data, the pooled paired t over 40 cells overstates precision (the pooled sign test inherits the same caveat); Section 6.2 therefore reports a supplementary, non-preregistered dataset-level check (the 8 dataset means as the unit, paired t and Wilcoxon) alongside the preregistered pooled test, and the conclusion rests on both.

5 What the arms actually subsample

Table 1 makes the degeneracy explicit. On all eight benchmark datasets, $q = 4P$ is at least the full off-diagonal pair count, so DiagonalFree-4P keeps every off-diagonal pair and drops only the diagonal self-products $e_i \odot e_i$, which duplicate the $e_p^2 - 1$ terms already present in the per-token block of Eq. (1). The preregistered primary result is therefore, literally, the statement "dropping the redundant diagonal self-products does not hurt" — plus the by-construction fact that the pair budget grows as $O(P)$ rather than $O(P^2)$. The $q = 2P$ arm is the only one that genuinely subsamples at these dimensions, keeping 50–80% of the off-diagonal pairs (100% on Airfoil, where $2P$ already covers all 10 pairs, making its two rows identical to the primary arm by construction — see the table note).

6 Results

6.1 Baseline fidelity

Our NumPy reimplementation is compared against the paper's reported Regression Block (RB) and FT-Transformer (FT-T) columns in Table 2. The reimplementation reproduces the RB column to within 0.008 per dataset and 0.001 on the mean, and recovers the exact 30,401 trainable parameters at $P = 8$. (FT-T is shown for reference; we did not reimplement it.)

Table 1: Per-dataset pair budgets. $q = 4P$ never subsamples the off-diagonal pair set at the benchmark dimensions: the primary arm tests diagonal redundancy, not sparsity. The $q = 2P$ arm is the true sparsity test. †On Airfoil ($P = 5$), $\min(2P, 10) = 10$, so the two arms select the identical 10 pairs from the same seeded RNG and their CSV rows coincide exactly — identical by construction, not a logging error.

Dataset	P	#off-diag. pairs	q=4P kept	q=2P kept
California	8	28	28 (100%)	16 (57%)
Yacht	6	15	15 (100%)	12 (80%)
Energy	8	28	28 (100%)	16 (57%)
Concrete	8	28	28 (100%)	16 (57%)
Airfoil†	5	10	10 (100%)	10 (100%)
Abalone	7	21	21 (100%)	14 (67%)
Kin8nm	8	28	28 (100%)	16 (57%)
Protein	9	36	36 (100%)	18 (50%)

Table 2: Out-of-sample R^2 (held-out variance explained): our reimplementation (mean ± SD over 5 splits) vs. the paper's reported FT-Transformer and Regression Block columns, with our DiagonalFree-4P and SparsePair-2P arms. The reimplementation matches the paper's RB column to 0.001 on the mean.

Dataset	full (ours)	DiagonalFree-4P	SparsePair-2P	FT-T	RB
California	0.767 ± 0.015	0.767 ± 0.013	0.764 ± 0.021	0.764	0.766
Yacht	0.992 ± 0.004	0.990 ± 0.004	0.986 ± 0.006	0.988	0.989
Energy	0.994 ± 0.002	0.995 ± 0.001	0.994 ± 0.001	0.994	0.995
Concrete	0.907 ± 0.013	0.903 ± 0.009	0.899 ± 0.005	0.898	0.907
Airfoil	0.919 ± 0.047	0.926 ± 0.026	0.926 ± 0.026	0.924	0.927
Abalone	0.332 ± 0.015	0.331 ± 0.014	0.336 ± 0.016	0.324	0.329
Kin8nm	0.922 ± 0.003	0.918 ± 0.003	0.906 ± 0.008	0.914	0.920
Protein	0.447 ± 0.026	0.447 ± 0.009	0.434 ± 0.023	0.426	0.455
Mean	0.785	0.785	0.781	0.779	0.786

6.2 Primary result: non-inferiority of the diagonal-free map

As Section 5 established, the preregistered primary arm keeps every off-diagonal pair — so this subsection tests, literally, whether "dropping the redundant diagonal self-products does not hurt." The preregistered primary arm never exercised random pair subsampling at any tested dimension; the subsampling mechanism itself is exercised only in the exploratory $q = 2P$ arm and the post-hoc follow-up. Per-dataset paired differences appear in Figure 1; Table 3 gives the pooled test alongside a dataset-level check. Across the 40 cells, DiagonalFree-4P minus full averages -0.0003; the 95% CI is $[-0.0049, +0.0043]$ (paired $t = -0.12$, $p = 0.906$; 16/40 wins, sign-test $p = 0.27$). The CI lower bound (-0.0049) lies well above the preregistered -0.01 margin: **non-inferiority holds**. Retrospectively, the pooled $n = 40$ design detects paired differences of ≈ 0.006 at 80% power (one-sided $\alpha = 0.05$)—the -0.01 margin is roughly 1.7× this detectable effect, so the verdict is not a power artifact; the $n = 3$ high-P follow-up remains exploratory as stated. The largest mean deficit on any dataset is -0.0041 (Concrete), under 0.005.

Because the five splits within a dataset share training data, the pooled test overstates precision. The pooled 40-cell mean also mixes heterogeneous scales — dataset-mean R^2 runs from 0.33 (Abalone) to 0.99 (Energy) — so a single pooled mean is a coarse summary and we keep the dataset-level $n = 8$ check prominent as the robustness anchor rather than leaning on the pooled test alone. A dataset-level analysis, taking the 8 dataset-mean differences as the unit to sidestep the split correlation, agrees: the mean is -0.0003 with 95% CI $[-0.0031, +0.0026]$ (paired $t = -0.23$, $p = 0.83$; Wilcoxon signed-rank $p = 0.64$) — and the conclusion rests on both.

The per-dataset 95% CIs in Table 4 ($n = 5$ each) are wide, and several cross the -0.01 margin: for the primary arm, Concrete ($[-0.0219, +0.0137]$), Airfoil ($[-0.0254, +0.0389]$), and Protein ($[-0.0385, +0.0379]$).

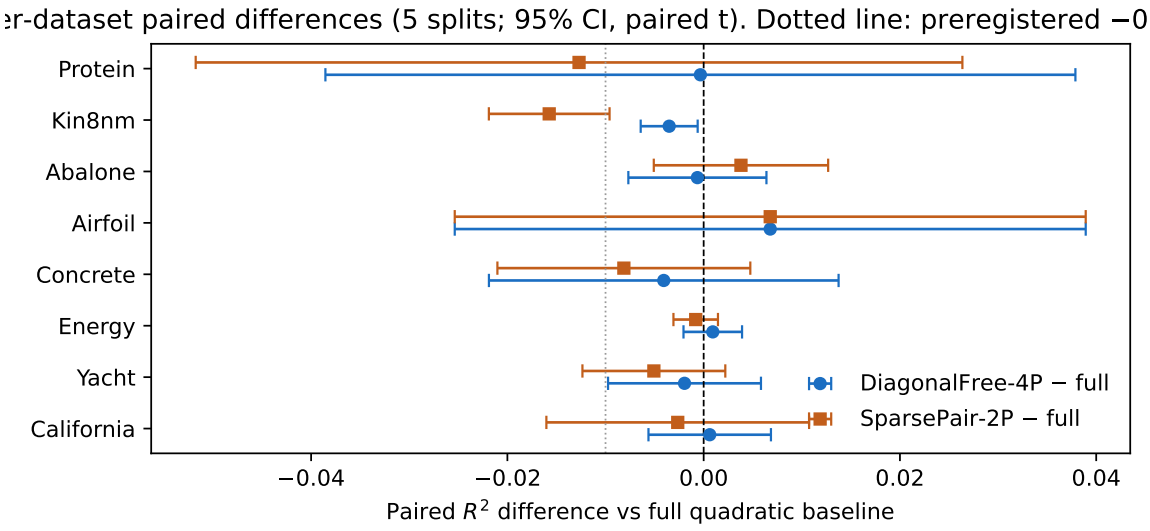


Figure 1: Per-dataset paired R^2 differences vs. the full quadratic baseline (5 splits each; 95% CI from a paired t with 4 df), dotted line: the preregistered -0.01 non-inferiority margin. Per-dataset CIs are wide and several cross the margin (see Table 4); the preregistered decision rule is the pooled test of Table 3, which clears the margin.

Table 3: Preregistered paired comparison over the 40 dataset×split cells, plus the dataset-level check ($n = 8$ dataset means, which sidesteps the within-dataset split correlation the pooled test ignores). DiagonalFree-4P is non-inferior to the full quadratic map at the 0.01 margin under both analyses. The SparsePair-2P arm is exploratory (not preregistered): its pooled CI sits inside the margin but its dataset-level CI marginally crosses it, and its losses are systematically negative (dataset-level 2/8, $p = 0.29$; pooled sign-test $p = 0.002$ across correlated splits). t: paired t-statistic; p: two-sided p-value; Wins: cells favoring the arm.

Contrast	Mean ΔR^2	95% CI	t	p	Wins
Pooled (40 cells)					
DiagonalFree-4P − full	−0.0003	[−0.0049, +0.0043]	−0.12	0.906	16/40
SparsePair-2P − full	−0.0043	[−0.0095, +0.0008]	−1.70	0.098	10/40
Dataset-level (8 means)					
DiagonalFree-4P − full	−0.0003	[−0.0031, +0.0026]	−0.23	0.83	3/8
SparsePair-2P − full	−0.0043	[−0.0108, +0.0022]	−1.58	0.159	2/8

That is the expected cost of $n = 5$ per-dataset inference, not evidence against the pooled finding — which is why the preregistered decision rule is the pooled test. A single split drives the Concrete width: split 4 gives $q4p - \text{full} = -0.0219$ while the other four range from -0.0091 to $+0.0166$, and we report it rather than trim it.

6.3 Genuine sparsity: the $q = 2P$ arm (exploratory)

In the preregistration, the $q = 2P$ arm was exploratory: no decision rule was specified, and we do not impose one post hoc here. It is nevertheless the only arm that genuinely sparsifies the pair set at these dimensions (Table 1), so it carries the paper's real sparsity evidence and deserves a preregistered-style accounting. Halving the pairs again — 50–80% of the off-diagonal pairs, 100% on Airfoil where the arms coincide by construction — costs 0.004 of mean R^2: 95% CI [−0.0095, +0.0008] pooled, [−0.0108, +0.0022] at the dataset level. Read against the -0.01 margin as a descriptive benchmark: the pooled CI sits inside

Table 4: Per-dataset paired 95% CIs for each arm minus the full baseline (5 splits, 4 df). Wide intervals are expected at $n = 5$; *marks CIs whose lower bound crosses the -0.01 margin. Airfoil's two rows coincide because the arms are identical there by construction (Table 1) — a single shared observation counted in both arms' tables, not two independent nulls.

Dataset	DiagonalFree-4P − full	SparsePair-2P − full
California	+0.0006 [−0.0056, +0.0069]	−0.0026 [−0.0160, +0.0107]*
Yacht	−0.0020 [−0.0097, +0.0058]	−0.0051 [−0.0124, +0.0022]*
Energy	+0.0009 [−0.0021, +0.0039]	−0.0008 [−0.0031, +0.0015]
Concrete	−0.0041 [−0.0219, +0.0137]*	−0.0081 [−0.0210, +0.0048]*
Airfoil	+0.0068 [−0.0254, +0.0389]*	+0.0068 [−0.0254, +0.0389]*
Abalone	−0.0006 [−0.0077, +0.0064]	+0.0038 [−0.0051, +0.0127]
Kin8nm	−0.0035 [−0.0064, −0.0006]	−0.0157 [−0.0219, −0.0096]*
Protein	−0.0003 [−0.0385, +0.0379]*	−0.0127 [−0.0518, +0.0264]*

Table 5: Paired head-to-head contrasts with the source's own PCA→50 trimming (40 cells; 95% CIs). Our Trimmed-50 keeps the feed-forward sublayer the source's Appendix E variant drops, so it is the PCA-trim component only. Trimming costs nothing measurable in our harness; against DiagonalFree-4P it is a tie on accuracy, while being cheaper in parameters and wall time. t: paired t-statistic; p: two-sided p-value; Wins: cells favoring the row's arm.

Contrast	Mean ΔR^2	95% CI	t	p	Wins
Trimmed-50 − full	−0.0000	[−0.0033, +0.0032]	−0.02	0.981	18/40
Trimmed-50 − DiagonalFree-4P	+0.0002	[−0.0031, +0.0036]	+0.14	0.890	18/40
SparsePair-2P − Trimmed-50	−0.0043	[−0.0084, −0.0001]	−2.09	0.043	18/40

it, the dataset-level CI marginally crosses it. These losses are not noise: 10/40 wins, systematically signed against SparsePair-2P (pooled sign-test $p = 0.002$ across correlated splits, reported with that caveat). At the dataset level the sign test gives 2/8 wins ($p = 0.29$), in line with the wide dataset-level CIs. They concentrate on Kin8nm (−0.016) and Protein (−0.013), while Abalone (+0.004) and Airfoil (+0.007) are unaffected or slightly improved. This pattern does not track the source paper's nonlinearity diagnostic: the OLS-vs-random-forest gaps in its Table 1 are widest on Yacht (0.417) and Airfoil (0.413) — both barely affected here — while Kin8nm (0.234) and Protein (0.236) sit mid-range. Aggressive sparsity bites dataset-specifically, not simply where nonlinearity is largest. We therefore do not claim non-inferiority for $q = 2P$; we report it as a measured ∼50%-sparsity operating point with a real cost signal — losses systematically negative (pooled sign-test $p = 0.002$ across correlated splits; dataset-level 2/8, $p = 0.29$) and concentrated on Kin8nm and Protein — rather than breaking down.

6.4 Head-to-head with the source's own trimming

The fair question the trimmed configuration raises is whether the source paper's own cost reduction — truncating the PCA basis 200 → 50 — beats ours. We ran it: the same 40 cells (same datasets, splits, and seeds) with the full quadratic pair map and PCA→50, differing from the full baseline in nothing else. One disclosure: our reimplementation keeps the feed-forward sublayer, whereas the source's Appendix E trimmed configuration also drops it; ours is the PCA-trim component only — the strict drop-in comparator, labeled Trimmed-50 throughout. Table 5 reports the paired contrasts, specified in the run script before any trim50 row existed — a post-hoc comparison, not part of the timestamped October 12 preregistration covering the main suite.

The outcome is a tie, and we report it as one. Truncating the PCA basis to 50 costs nothing measurable in our harness: mean $\Delta R^2 = -0.0000$ vs. the full block (95% CI [−0.0033, +0.0032], 18/40 wins) — an independent confirmation, on identical seeds, of the source paper's own claim that the 50-PC trim is within noise at $P \leq 9$. Against DiagonalFree-4P, the two cost reductions are indistinguishable on accuracy: +0.0002

([−0.0031, +0.0036], 18/40 wins). Only the exploratory $q = 2P$ arm loses to the trimming, by −0.0043 ([−0.0084, −0.0001]) — the same magnitude it loses to the full block. That $p = 0.043$ is one of many uncorrected contrasts we computed, so we read it as weak evidence rather than a discovered effect.

Read plainly, at $P \leq 9$ the source's own trimming dominates on every *measured* cost: 20,801 trainable parameters against our deliberately unchanged 30,401, and mean wall time 23 s against 89 s for DiagonalFree-4P (training is early-stopping-dominated; ours saves nothing measurable at these dimensions). What SparsePair-2P buys that trimming cannot is the construction-cost scaling itself: $O(P)$ pairs instead of $O(P^2)$, which only bites at larger P (Section 6.5). The two are different points on the trade-off curve, and now both are measured.

6.5 Higher-dimension follow-up: $P = 21$ and $P = 32$ (post-hoc)

The preregistered suite never exercises the $O(P)$ scaling it was designed around, so we ran a post-hoc follow-up — same protocol family, not preregistered, reported descriptively — on two datasets from the source paper's own Appendix F higher-dimension suite: CPU_act ($P = 21$, $N = 8192$) and Bank32nh ($P = 32$, $N = 8192$), N capped at 5000 as in the source, three splits (seeds 42, 1042, 2042), full baseline vs. DiagonalFree-4P. Here $q = 4P$ genuinely subsamples: 84 of 210 off-diagonal pairs (40%) at $P = 21$, 128 of 496 (26%) at $P = 32$. At these dimensions, the full-batch PCA fit exceeds this VM's RAM (∼21 GB at $P = 32$), so randomized SVD fits the PCA on a seeded subsample of at most 10,000 training tokens — the one documented deviation, and itself evidence for the construction-cost claim: the full-batch PCA fit is over the quadratic-width feature map, so the subsampling the RAM ceiling forces is the construction bottleneck, not a downstream modeling choice. The full baselines nevertheless reproduce the source's own numbers on these datasets (0.973 and 0.509 in its Table 5 Panel B, against our 0.9707 and 0.5087), so the subsampled-PCA protocol leaves the baseline credible. The paired comparison remains internal, so it still isolates the feature map. We chose CPU_act and Bank32nh as the two smallest of the source's higher-dimension suite — a convenience choice for CPU feasibility that biases toward the least aggressive subsampling dimensions, so it cannot trace the sparsity-limit curve at more aggressive ratios (e.g. $P = 48$).

Table 6: Higher-dimension follow-up (post-hoc, 3 splits each): full baseline vs. DiagonalFree-4P on CPU_act ($P = 21$) and Bank32nh ($P = 32$). $q = 4P$ genuinely subsamples here (84/210 and 128/496 off-diagonal pairs). Accuracy stays close to the baseline while the constructed feature width falls 63% and 75%. No preregistered decision rule applies; results are descriptive.

Dataset	full R^2	DiagonalFree-4P R^2	mean diff.	pairs kept	width (features)
CPU_act (P=21)	0.9707	0.9713	+0.0006 [−0.0013, +0.0025]	84/210 (40%)	14,912→5,504
Bank32nh (P=32)	0.5087	0.5049	−0.0039 [−0.0160, +0.0082]	128/496 (26%)	33,920→8,320

On CPU_act DiagonalFree-4P is descriptively close to the full baseline: +0.0006 on the mean, 95% CI [−0.0013, +0.0025] ($n = 3$). On Bank32nh it gives up 0.0039 on the mean (CI [−0.0160, +0.0082], which crosses the −0.01 margin widely); each dataset's mean sits within about 0.007 of the baseline (Table 6). With three splits the follow-up is underpowered to test non-inferiority, so we report this as descriptive closeness, not a second "match." The direct per-batch measure of the construction cost, the constructed feature width, falls from 14,912 to 5,504 at $P = 21$ and from 33,920 to 8,320 at $P = 32$ — at identical trainable parameter counts (31,233 and 31,937). Per-run wall time remains early-stopping-dominated (as at $P \leq 9$), so we report the width reduction rather than a wall-clock speedup. This is the evidence the preregistered suite could not provide: genuine subsampling, at the paper's stated pain-point dimensions, with descriptively reported accuracy differences.

6.6 Cost and parameters

To state it without ambiguity: the construction-cost claim in this paper is width-based and analytic, not a measured time-or-memory speedup. Trainable parameters are identical across methods (30,401 at $P = 8$): the PCA still outputs 200 components and the $200 \to 64$ mixer is untouched. The construction-cost saving

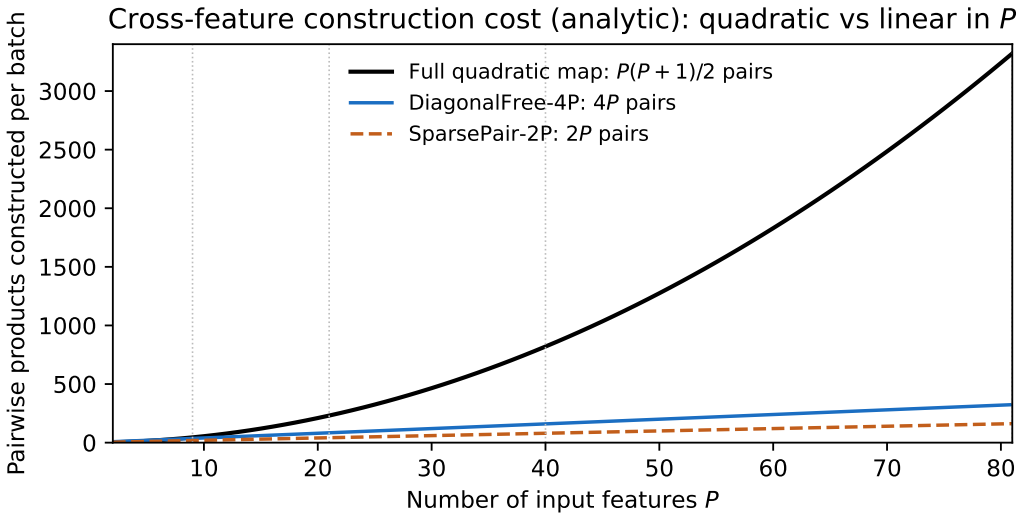


Figure 2: Pairwise products constructed per batch as a function of input dimension P: quadratic for the full map, linear for the sparse variants. Analytic plot of the pair-count formulas, not measured timings. Dotted lines mark $P = 9$ (preregistered suite max), $P = 21$ and 32 (post-hoc follow-up), $P = 40$ (an intermediate dimension), and $P = 81$ (the source's reported binding-constraint regime: the author's runs there required an A100 GPU and one timed out).

shows up nowhere in the parameter count, and we do not present the identical count as a measured benefit. What changes is the constructed feature width: at $P = 8$, 2,432 (full), 1,920 (DiagonalFree-4P), 1,152 (SparsePair-2P). Figure 2 shows the asymptotic construction cost: $P(P+1)/2$ vs. $4P$ vs. $2P$ pair products per batch — an analytic plot of the pair-count formulas, not measured timings. Mean wall time was ≈90 s per run on a 2-CPU VM for all methods, and several sparse runs were slower than their baselines (e.g. Concrete split 0: 70 s vs. 24 s); per-run timing at these small P is dominated by early-stopping epochs rather than the map. We measured no wall-clock or memory benefit at $P \leq 9$ and claim none there; the measured construction-cost win is the width reduction at larger P (Section 6.5).

7 Discussion and limitations

Stated without the compression that abstracts invite, the honest headline is this: the preregistered primary test shows the Regression Block's diagonal self-products are redundant. Dropping them costs nothing measurable on the paper's full benchmark suite — pooled mean −0.0003 (95% CI [−0.0049, +0.0043]), dataset-level CI [−0.0031, +0.0026]. The exception is Kin8nm, where the primary arm sits consistently below the baseline (−0.0035 mean, per-dataset CI entirely below zero) — a reminder that "redundant" is a pooled statement. With eight uncorrected per-dataset CIs, the chance that at least one sits entirely below zero at the 5% level by noise alone is $P \approx 0.18$ (one-sided, $1 - 0.975^8$), so we read the Kin8nm exception as plausible noise, not a discovered dataset-specific effect. The $q = 2P$ arm, the only one that genuinely sparsifies at these dimensions, keeps 50–80% of the off-diagonal pairs at a measured cost of ≈0.004 mean R^2, with systematically negative losses (dataset-level 2/8, $p = 0.29$; pooled sign-test $p = 0.002$ across correlated splits) concentrated where interactions carry the most signal. The $O(P)$ construction scaling is by design; its empirical bite is at larger P.

On the "corollary" objection: the source paper's 50-PC ablation showed that the *post-hoc PCA width* is redundant (50 components retain > 99.99% of the polynomial basis's variance), not that the *construction* of the $O(P^2)$ pair products is. Those are different claims: one can compress a basis after paying to build it, as the paper's trimmed configuration does. We instead test whether the pairs need building at all, and propose and test a fix — which the source paper does not. The trimmed configuration is also the honest comparator at

the pain point. At $P = 21$–48 it loses 0.010–0.036 relative to the full Regression Block (source Appendix F). At $P \leq 9$ we measured the PCA-trim component head-to-head in our own harness (Section 6.4): truncating $200 \to 50$ costs -0.0000 (95% CI $[-0.0033, +0.0032]$) against the full block on identical seeds — the PCA width itself is free, independently confirming the source's "within 0.002" claim, and suggesting the 0.011 deficit the source reports for its fuller no-feed-forward trim lives in the dropped sublayer rather than the basis width. Read plainly, at these dimensions the source's own trimming dominates on every measured cost (20,801 parameters and 23 s mean wall time against our deliberately unchanged 30,401 and 89 s). Our variant is a different trade: a strict drop-in that changes no parameter, sublayer, or training choice, targeting the construction cost the trimmed variant leaves untouched.

The higher-dimension follow-up (Section 6.5) supplies the empirical bite the preregistered suite lacks. At $P = 21$ and $P = 32$ — where $q = 4P$ keeps only 40% and 26% of the off-diagonal pairs — DiagonalFree-4P is descriptively close to the full baseline ($+0.0006$ and -0.0039 mean ΔR^2 over three splits each, but the Bank32nh CI $[-0.0160, +0.0082]$ crosses the -0.01 margin widely: with $n = 3$ this follow-up is underpowered to test non-inferiority) while the constructed feature width falls 63% and 75%. Two caveats keep this in its place. First, it is post-hoc and unpreregistered: three splits, not five, and a subsampled randomized PCA fit forced by this VM's RAM, so it is supporting evidence, not a second preregistered test. Second, the source paper's trimmed configuration remains the cheaper model in absolute parameters (7,489 at $P = 8$ in the source's Table 4 — a different counting scope than the 20,801 measured in our harness for the trimmed variant; against our deliberately unchanged count); what ours buys is the construction-cost scaling without touching the architecture, the parameters, or the training — a different point on the trade-off curve, and the one the source paper left untested.

Limitations. First, the preregistered suite runs entirely at $P \leq 9$, where the full quadratic map is CPU-trivial; the pain-point regime is covered only by the post-hoc follow-up above. Second, this is a NumPy reimplementation rather than the author's torch code. Exact numeric replication was never expected; ballpark agreement was achieved (Section 6.1), and the paired test is internal — which is the valid comparison. Third, five splits per dataset give modest power and wide per-dataset CIs (Table 4). The pooled CI (± 0.005) sits well inside the margin, so the primary conclusion is not power-limited; but the $q = 2P$ arm's dataset-level CI ($[-0.0108, +0.0022]$) should be read with that modesty in mind. The $q = 2P$ pair subset is a single fixed draw per split from a seeded RNG, and sensitivity to the subset draw itself is entirely untested: a different random draw could shift the loss magnitude or its dataset concentration. In particular, the Kin8nm losses at $q = 2P$ are consistently signed across all five splits (mean -0.016, per-dataset CI entirely below zero) — a systematic pattern that draw variation, not just the sparsity level, could plausibly explain, and that a single-draw-per-split design cannot disentangle from sparsity. We did not reimplement the FT-Transformer; its column is reference only.

8 Related work

The Regression Block and its evaluation are due to Goulet Coulombe (2026), built on the FT-Transformer skeleton of Gorishniy et al. (2021); the OLS-as-attention equivalence itself builds on the transformer architecture of (Vaswani et al., 2017). Sparsifying interactions is an old idea in new clothes: neural additive models restrict themselves to univariate and pairwise shape functions (Agarwal et al., 2021), and random-feature maps approximate polynomial kernels without forming the explicit $O(P^2)$ expansion (Kar & Karnick, 2012). Polynomial sketching methods such as TensorSketch likewise avoid materializing the quadratic expansion, projection into a compact sketch at near-linear construction cost (Pham & Pagh, 2013). Sketching is the stronger remedy on paper: TensorSketch avoids materializing the quadratic expansion while preserving approximation guarantees (variance $\sim 1/m$ for sketch size m, with a subspace-embedding guarantee). Pair subsampling is, by contrast, the weakest sketch — a subsampled identity with variance $\sim P^2/q$ and no such guarantee — and we test it precisely because it is the cheapest unstructured alternative in implementation effort, not because it is theoretically competitive; we measured no wall-clock benefit at any P. One directly relevant quantity we do not measure is coherence: uniform subsampling degrades exactly when the quadratic features are coherent, which is consistent with the losses concentrating on Kin8nm and Protein (Section 6.3) and makes coherence-aware sampling the obvious untested alternative. No sketching construction has been tested as a drop-in for the Regression Block's mixer; this paper tests the weaker alternative instead. Closest

in spirit, TabPFN offers second-order polynomial features over up to 50 randomly selected feature pairs as a preprocessing option (Hollmann et al., 2025). The source paper itself reports the closest cost experiment — the trimmed configuration of its Appendix E (two mixing sublayers, no feed-forward sublayer, 50 principal components). At $P = 21$–48 it loses 0.010–0.036 relative to the full Regression Block and 0.012–0.062 relative to the FT-Transformer (Goulet Coulombe, 2026). That experiment cuts the PCA basis width and the feed-forward sublayer, presented as an ablation rather than a proposed fix, and leaves the $O(P^2)$ construction untouched. None of the above targets the construction cost of the Regression Block's mixer, and none comes with a preregistered parity test; that is the gap this paper fills.

9 Conclusion

We preregistered a pair-subsampling replacement for the Regression Block's quadratic cross-feature map — a fixed subset of feature pairs, with architecture, parameters, and training untouched. At benchmark dimensions, however, the preregistered $q = 4P$ rule retains every off-diagonal pair and drops only the diagonal self-products: the primary test is a diagonal ablation, not a subsampling test, and we report it as one. On the paper's eight benchmarks $\times$ five splits, the diagonal ablation is non-inferior to the full map. At these dimensions, it retains every off-diagonal pair and drops only the redundant diagonal self-products; the mean ΔR^2 is -0.0003 (95% CI $[-0.0049, +0.0043]$) against a 0.01 margin, and the dataset-level check agrees. The exploratory $q = 2P$ arm, the true $\sim$50%-sparsity test at these dimensions, costs $\approx$0.004 mean R^2, with systematically negative losses (dataset-level 2/8 wins, $p = 0.29$; pooled sign-test $p = 0.002$ across correlated splits) concentrated on Kin8nm and Protein — though that concentration does not track the source's nonlinearity diagnostic. In a post-hoc follow-up at $P = 21$ and $P = 32$, where $q = 4P$ genuinely subsamples the pair set, accuracy is descriptively close to the baseline — mean ΔR^2 $+0.0006$ and -0.0039, all six cells within about 0.007, though the Bank32nh CI ($[-0.0160, +0.0082]$) crosses the -0.01 margin widely — while the constructed feature width falls 63% and 75%. At $n = 3$ this follow-up is underpowered to test non-inferiority, so it is reported descriptively, not as a "match." What the experiments do not show is a measured wall-clock benefit at any P — per-run timing is early-stopping-dominated — so the construction-cost claim rests on the measured width reduction (63–75% at $P = 21/32$) and the analytic $O(P^2) \to O(P)$ pair-count scaling beyond that.

Statement of AI Use

The authors used the AI tool Muse (Muse Spark 1.3, Meta) for drafting, experiment design, code implementation and validation, and data analysis. The contents are reviewed, edited, and attested for accuracy by the authors.

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A Preregistration record

The preregistration document (timestamped 2026-10-12, before any run) specified: the DiagonalFree-4P primary variant and SparsePair-2P exploratory variant with the random-pair mechanism of Section 3; the eight datasets with no exclusions; five 80/20 splits at seeds $42 + 1000k$ with the paper's $N = 5000$ cap and target standardization; out-of-sample R^2; the non-inferiority margin (-0.01) and decision rule of Section 4; and the smoke-test gate (baseline in the paper's ballpark before scaling). No deviations occurred. All 120 preregistered runs completed with status ok; we recomputed every number in the main suite from `outputs/results.csv`, and the higher-P follow-up numbers from `outputs/results_highp.csv` (Section 6.5).

One *post-hoc* addition, not covered by the preregistration and reported as such: a higher-dimension follow-up (Section 6.5) running the same protocol family at $P = 21$ and $P = 32$ — the source paper's Appendix F regime, where $q = 4P$ genuinely subsamples the pair set — with the PCA fit done as a randomized-SVD fit (sklearn `PCA(svd_solver='randomized')`) on a seeded subsample of at most 10,000 training tokens for memory reasons (the code's actual route; a batched IncrementalPCA attempt was abandoned as too slow). No preregistered decision rule applies to it; its results are reported descriptively.

B Implementation notes

NumPy reimplementation of the paper's Regression Block following its Appendix C and the MIT-licensed reference implementation's Table-1 pipeline (adapted with attribution; no torch). Backpropagation was verified by float64 finite differences (all 15 parameter groups, relative error $< 10^{-4}$). Determinism: rerunning any cell reproduces its R^2 to six decimals (all RNG streams seeded). One environment note: the VM's default NumPy linked the reference BLAS (≈90 MFLOPS on the PCA matmuls); we installed the PyPI NumPy 1.26.4 wheel (OpenBLAS) into user space, same version, which the timing figures reflect. sklearn issued `LinAlgWarning`s on some ridge warm-start fits (ill-conditioned); runs completed normally, as in the reference code path.

C Per-cell results

Full per-split R^2 values are in `outputs/results.csv` (120 rows) for the main suite and `outputs/results_highp.csv` for the higher-P follow-up. Dataset means and SDs are in Table 2; paired differences per dataset are plotted in Figure 1.

Data and Code Availability

All analysis code, per-split results (`outputs/results.csv`, `outputs/results_highp.csv`), and the reproduction package are included as supplementary material. A public code repository will be released upon acceptance.

Competing Interests

The author declares no competing interests. No funding was received for this work.